

PSET 6: Discrete random variables

Assignment Qs

- Name
- How long did this problem set take you (round to nearest hour)? 0-1 hour 2-3 hours 4-5 hours 5+ hours
- How difficult was this problem set? very easy 1 2 3 4 5 very challenging

Problem 1: Calculate probabilities in a sample space S

Events A and B are contained within a sample space S . Given that $\Pr(A) = 0.55$, $\Pr(B) = 0.3$, and $\Pr(A \cap B) = 0.15$, find:¹

- $\Pr(A \cup B)$
- $\Pr(A \cap B^c)$
- $\Pr[(A \cap B^c) \cup (B \cap A^c)]$
- Are A and B independent? Are they mutually exclusive? Explain how these concepts differ.

Problem 2: Random variables

Provide the following explanations in your own words:

- What is a random variable?
- What is the difference between upper case and lower case x ? How (if at all) do they matter?
- If I'm thinking about something like $p_X(x_0)$, what do all the parts/pieces mean?
- What is a discrete random variable?

Problem 3: Survey says

A survey has 54% respondents 50 or older and 46% respondents under 50. Within the survey, on a particular question, 9.5% of the 50-plus population agrees strongly while 2.7% of under-50 respondents agree strongly.

- What is the probability someone selected at random is 50 or older?

¹Inspired by Grimmer HW8.4

- b. The selected individual strongly agrees with the survey question. Now what is the likelihood that person is 50 or older? Explain your reasoning and show all your work.
- c. Are the two answers above the same or different? Explain.
- d. (for fun, no points) What is the survey question?

Problem 4: PMF vs. CMF

Consider the following function: $f(x) = \frac{1}{6}$.

- a. Find the pmf and cmf of the function and provide them in a table below. *You may need to make some assumptions for this to work – clearly state any assumptions you make*
- b. What is the variance of X ?
- c. Now consider a different function, $p(x) = cx$ for $x \in \{1, 2, 3, 4\}$ and 0 otherwise. For what value of c is this a valid pmf? State which property of a pmf you used to find it.
- d. Using the value of c you found, calculate $E[X]$ and $\text{Var}(X)$.

Problem 5: Building a distribution from scratch

Three countries are independently deciding whether to sign a treaty. Each country signs with probability 0.6. Let X be the number of countries that sign.

- a. Write out the sample space. How many outcomes does it contain?
- b. Construct the pmf of X . Show how you collected outcomes together to obtain $p_X(x)$ for each value of x .
- c. Verify that your pmf satisfies the properties of a valid pmf.
- d. Construct the cmf of X and present it in a table alongside the pmf.
- e. Find $E[X]$ and $\text{Var}(X)$.
- f. Is X a named distribution? If so, which one, and with what parameters? Do your answers to (e) match the formulas for that distribution's mean and variance?

Problem 6: Getting a traffic ticket

You drive to work 5 days a week for 10 weeks, and with probability $p = 0.06$ you get a traffic ticket on any given day, independent of other days. Let X be the total number of tickets you get in this time.²

- a. What is the probability that the number of tickets you get is exactly equal to the expected value of X ?

²Inspired by BT 2.41

- b. Calculate approximately the probability in (a) using a Poisson approximation.
- c. Now suppose you are instead interested in how long you can go before your first ticket. What is the probability that your first ticket comes on the 10th day you drive?
- d. What is the expected number of days you drive until you get your first ticket? Which distribution did you use here, and why is it a different one from the distribution you used in (a)?

Problem 7: Obtaining requests for information

X is a discrete random variable. It takes the value of the number of days required for a governmental agency to respond to a request for information. X is distributed according to the following PMF:³

$$f(x) = e^{-6} \frac{6^x}{x!} \quad \text{for } x \in \{0, 1, 2, \dots\}$$

- a. Given this information, what is the probability of a response from the agency in 5 days or less?
- b. What is the probability the agency response takes more than 10 but less than 13 days?
- c. What is the probability the agency response takes more than 5 days?
- d. Suppose using X you generate a new variable, **Responsive**. **Responsive** equals 1 if an agency responds in 5 days or less and 0 otherwise. What is the expected value of **Responsive**?
- e. What is the variance of **Responsive**?
- f. The agency charges a processing fee for each request: a flat \$25.00, plus \$10.00 for every day it takes them to respond. Let $C = 25 + 10X$ be the fee you are charged. Find $E[C]$ and $\text{Var}(C)$. **Hint: you should not need to recompute anything from scratch.**

Problem 8: Modeling electoral outcomes

Suppose we've developed a model predicting the outcome of the upcoming midterm elections in a state with 4 Congressional districts. In each district there are two candidates, a Republican and a Democrat. We have reason to believe the following PMF describes the distribution of potential election results, where $K \in \{0, 1, 2, 3, 4\}$ is the number of seats won by Republican candidates in the upcoming election.

$$\Pr(K = k \mid \theta) = \binom{4}{k} \theta^k (1 - \theta)^{4-k}$$

Based on polling information, we think the appropriate value for θ is 0.423.⁴

- a. How many distinct ways can Republicans win exactly 2 of the 4 seats?
- b. What's the expected number of seats Republicans will win in the upcoming election?

³Inspired by Grimmer HW10.2

⁴Inspired by Grimmer HW10.3. Data from <https://www.realclearpolling.com/polls/state-of-the-union/generic-congressional-vote>

- c. Given this PMF, what's the probability that no Republican legislators win in the upcoming election?
- d. What's the probability that Republican legislators win a majority of the seats in this state?
- e. A prominent political pundit declares they are certain that Republicans will win a majority of seats in the next election and offers the following bet. If Republicans win a majority of the seats, we must pay the pundit \$15.00. If Republicans fail to win a majority of the seats, we will win \$20.00. Based on our model, should we take this bet? **Hint: think of the betting outcomes as a random variable. Find the expected value of this random variable.**
- f. Suppose we are offered a second bet with a more complicated structure. In this case we'll receive \$100 if the Republicans win a majority, \$50 if neither party wins a majority, and we'll have to pay \$200 if the Democrats win a majority. Should we take this bet?

AI and Resources statement

Please list (in detail) all resources you used for this assignment. If you worked with people, list them here as well. It is not enough to say that you used a resource for help, you need to be specific on the link and *how* it was helpful. W/R/T gen AI tools (including GPT, Claude, etc.) you cannot use them to do work on your behalf—you cannot put in any of the questions, etc. You can ask for help on logic / sample problems. If you do use GPT or other AI tools, you need to provide a link to your chat transcript. Any suspected academic integrity violations will be immediately reported to the Dean of Students.